

DSC 206: Homework 4

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Assignment 1 Consider the data stream 1, 2, 6, 3, 2, 2, 3, 1, 6, 6, 5, 4, 2, 5. Choose $d = 2$ and $w = 5$ to compute the Count-min sketch as described in class, using the two hash functions $h_1(x) = (x + 2) \bmod 5$ and $h_2(x) = (2x + 1) \bmod 5$.

- (a) Construct the $d \times w$ hash table produced by the Count-min sketch algorithm.
- (b) Return the estimated frequency of each item using the Count-min sketch.

Solution. (a) The distinct items are $\{1, 2, 3, 4, 5, 6\}$ with true counts

$$c(1) = 2, \quad c(2) = 4, \quad c(3) = 2, \quad c(4) = 1, \quad c(5) = 2, \quad c(6) = 3.$$

Each item increments one cell per row, at the columns given by the hash functions:

x	1	2	3	4	5	6
$h_1(x) = (x + 2) \bmod 5$	3	4	0	1	2	3
$h_2(x) = (2x + 1) \bmod 5$	3	0	2	4	1	3

Adding each item's true count into the cell it hashes to (note that in *both* rows the items 1 and 6 collide in column 3, since $h_1(1) = h_1(6) = 3$ and $h_2(1) = h_2(6) = 3$) gives the 2×5 sketch:

	col 0	col 1	col 2	col 3	col 4
row h_1	2	1	2	5	4
row h_2	4	2	2	5	1

Row h_1 : col 0 holds 3; col 1 holds 4; col 2 holds 5; col 3 holds 1+6; col 4 holds 2. Row h_2 : col 0 holds 2; col 1 holds 5; col 2 holds 3; col 3 holds 1+6; col 4 holds 4.

(b) The Count-min estimate is $\hat{c}(x) = \min\{T[1, h_1(x)], T[2, h_2(x)]\}$:

x	$h_1(x)$	$T[1, \cdot]$	$h_2(x)$	$T[2, \cdot]$	$\hat{c}(x) = \min$	true $c(x)$
1	3	5	3	5	5	2
2	4	4	0	4	4	4
3	0	2	2	2	2	2
4	1	1	4	1	1	1
5	2	2	1	2	2	2
6	3	5	3	5	5	3

The estimates for 2, 3, 4, 5 are exact. The estimates for 1 and 6 are both 5, an overestimate: these two items collide in *both* rows ($h_1(1) = h_1(6)$ and $h_2(1) = h_2(6)$), so no row can separate them and each inherits $c(1) + c(6) = 2 + 3 = 5$. This illustrates that Count-min never underestimates ($\hat{c}(x) \geq c(x)$) and only overestimates when collisions occur in every row.

Assignment 2 Consider running the (online) Perceptron algorithm on a sequence of examples S . Let S' be the same set of examples presented in a different order. Does the Perceptron algorithm necessarily make the same number of mistakes on S as on S' ? If so, why; if not, exhibit such S, S' on which the mistake counts differ.

Solution. No, the number of mistakes is *not* order-invariant. While the classical mistake bound R^2/γ^2 depends only on the geometry of the point set (and so is the same for any ordering), the *realized* number of mistakes can change with the order, because each mistake updates w and thereby changes which later points are misclassified.

Convention. The algorithm starts at $w = \mathbf{0}$, predicts $\hat{y} = \text{sign}(w \cdot x)$, and makes a mistake whenever $\text{sign}(w \cdot x) \neq y$; on the first example $w = \mathbf{0}$ gives $w \cdot x = 0$, which is counted as a mistake (the zero vector cannot classify any point). On a mistake it updates $w \leftarrow w + yx$.

Counterexample. Take the three points in $\mathbb{R}^2$

$$p_1 = (2, -1), y_1 = +1; \quad p_2 = (-1, 2), y_2 = +1; \quad p_3 = (-1, -1), y_3 = -1.$$

They are linearly separable: $w^* = (1, 1)$ gives $w^* \cdot p_1 = 1 > 0$, $w^* \cdot p_2 = 1 > 0$, $w^* \cdot p_3 = -2 < 0$.

Order $S = (p_1, p_2, p_3)$:

- p_1 : $w = \mathbf{0}$, $w \cdot p_1 = 0 \Rightarrow$ **mistake**; update $w = (0, 0) + (+1)(2, -1) = (2, -1)$.
- p_2 : $w \cdot p_2 = (2, -1) \cdot (-1, 2) = -4 < 0$, predict -1 but $y = +1 \Rightarrow$ **mistake**; update $w = (2, -1) + (-1, 2) = (1, 1)$.
- p_3 : $w \cdot p_3 = (1, 1) \cdot (-1, -1) = -2 < 0$, predict $-1 = y \Rightarrow$ **correct**.

Total: $\boxed{2}$ mistakes.

Order $S' = (p_3, p_1, p_2)$ (same set, reordered):

- p_3 : $w = \mathbf{0}$, $w \cdot p_3 = 0 \Rightarrow$ **mistake**; update $w = (0, 0) + (-1)(-1, -1) = (1, 1)$.
- p_1 : $w \cdot p_1 = (1, 1) \cdot (2, -1) = 1 > 0 =$ predict $+1 = y \Rightarrow$ correct.
- p_2 : $w \cdot p_2 = (1, 1) \cdot (-1, 2) = 1 > 0$, predict $+1 = y \Rightarrow$ correct.

Total: $\boxed{1}$ mistake.

Thus the same examples in a different order yield 2 versus 1 mistakes. Intuitively, starting with the negative example p_3 immediately drives w to the separator $(1, 1)$, whereas starting with p_1 first pushes w to $(2, -1)$, which misclassifies p_2 and forces an extra correction.

Assignment 3 Suppose we modify the (batch) Perceptron so that on each mistake we update $w \leftarrow w + \eta \text{Label}(x_i) x_i$ for some fixed $\eta > 0$ (instead of $\eta = 1$), where $\text{Label}(x) \in \{+1, -1\}$. Assuming the data are linearly separable, derive an upper bound on the number of updates this algorithm makes. Provide full details.

Solution. Setup. Since the data are linearly separable, there is a unit vector w^* ($\|w^*\| = 1$) and a margin $\gamma > 0$ with

$$y_i (w^* \cdot x_i) \geq \gamma \quad \text{for all examples } (x_i, y_i),$$

where $y_i = \text{Label}(x_i)$. Let $R = \max_i \|x_i\|$. Start from $w_0 = \mathbf{0}$, and let M be the total number of updates. We bound $w_M \cdot w^*$ from below and $\|w_M\|$ from above, then combine.

Step 1: lower bound on the projection onto w^* . On an update caused by example (x_i, y_i) ,

$$w_{k+1} \cdot w^* = (w_k + \eta y_i x_i) \cdot w^* = w_k \cdot w^* + \eta y_i (x_i \cdot w^*) \geq w_k \cdot w^* + \eta \gamma.$$

Summing over the M updates (and $w_0 \cdot w^* = 0$),

$$w_M \cdot w^* \geq M \eta \gamma. \tag{1}$$

Step 2: upper bound on the norm. A mistake on (x_i, y_i) means $y_i (w_k \cdot x_i) \leq 0$. Hence

$$\|w_{k+1}\|^2 = \|w_k + \eta y_i x_i\|^2 = \|w_k\|^2 + 2\eta y_i (w_k \cdot x_i) + \eta^2 \|x_i\|^2 \leq \|w_k\|^2 + 0 + \eta^2 R^2.$$

Summing over the M updates (and $\|w_0\|^2 = 0$),

$$\|w_M\|^2 \leq M \eta^2 R^2, \quad \text{i.e.} \quad \|w_M\| \leq \eta R \sqrt{M}. \tag{2}$$

Step 3: combine via Cauchy–Schwarz. Since $\|w^*\| = 1$,

$$w_M \cdot w^* \leq \|w_M\| \|w^*\| = \|w_M\| \leq \eta R \sqrt{M}.$$

Chaining with (1),

$$M \eta \gamma \leq \eta R \sqrt{M} \implies \sqrt{M} \gamma \leq R \implies \boxed{M \leq \frac{R^2}{\gamma^2}}.$$

Key insight. The learning rate η scales the lower bound (1) linearly and the upper bound (2) by the *same* factor η (after taking the square root), so η cancels exactly. The mistake bound R^2/γ^2 is therefore *independent of η* : a constant step size only rescales the iterate w and does not change the geometry of the problem.

Assignment 4 For each function, decide whether it is convex and prove your claim.

(a) $f(\mathbf{x}) = x_1^2 - 2x_2^2 + 3$, where $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$.

(b) Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$ be convex, and $A \in \mathbb{R}^{D \times d}$. Define $g : \mathbb{R}^d \rightarrow \mathbb{R}$ by $g(\mathbf{y}) = f(A\mathbf{y})$.

Decide whether g is convex. (Prove directly from the definition; no theorems from class.)

(c) $F(\mathbf{x}) = g(e^{x_1} + e^{x_2})$, where $\mathbf{x} \in \mathbb{R}^2$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ is monotonically increasing and convex.

Solution. (a) **Not convex.** The Hessian is $\nabla^2 f = \begin{pmatrix} 2 & 0 \\ 0 & -4 \end{pmatrix}$, which has a negative eigenvalue -4 , so f is not convex. A concrete violation of the definition: take $\mathbf{u} = (0, 1)$, $\mathbf{v} = (0, -1)$, $\alpha = \frac{1}{2}$. Then $\frac{1}{2}\mathbf{u} + \frac{1}{2}\mathbf{v} = (0, 0)$ and

$$f\left(\frac{1}{2}\mathbf{u} + \frac{1}{2}\mathbf{v}\right) = f(0, 0) = 3, \quad \frac{1}{2}f(\mathbf{u}) + \frac{1}{2}f(\mathbf{v}) = \frac{1}{2}(1) + \frac{1}{2}(1) = 1.$$

Since $3 > 1$, convexity ($f(\text{midpoint}) \leq \text{average}$) fails.

(b) **g is convex.** Fix $\mathbf{y}_1, \mathbf{y}_2 \in \mathbb{R}^d$ and $\alpha \in [0, 1]$. By linearity of A ,

$$g(\alpha\mathbf{y}_1 + (1 - \alpha)\mathbf{y}_2) = f(A(\alpha\mathbf{y}_1 + (1 - \alpha)\mathbf{y}_2)) = f(\alpha A\mathbf{y}_1 + (1 - \alpha) A\mathbf{y}_2).$$

Set $\mathbf{u} = A\mathbf{y}_1$, $\mathbf{v} = A\mathbf{y}_2 \in \mathbb{R}^D$. By convexity of f ,

$$f(\alpha\mathbf{u} + (1 - \alpha)\mathbf{v}) \leq \alpha f(\mathbf{u}) + (1 - \alpha)f(\mathbf{v}) = \alpha f(A\mathbf{y}_1) + (1 - \alpha)f(A\mathbf{y}_2).$$

Hence

$$g(\alpha\mathbf{y}_1 + (1 - \alpha)\mathbf{y}_2) \leq \alpha g(\mathbf{y}_1) + (1 - \alpha)g(\mathbf{y}_2),$$

so g is convex. (Convexity is preserved under composition with an affine/linear map.)

(c) **F is convex.** Write $h(\mathbf{x}) = e^{x_1} + e^{x_2}$. Fix $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ and $\alpha \in [0, 1]$.

Step 1 (h is convex). Using convexity of $t \mapsto e^t$ coordinatewise,

$$h(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}) = e^{\alpha x_1 + (1 - \alpha)y_1} + e^{\alpha x_2 + (1 - \alpha)y_2} \leq \alpha e^{x_1} + (1 - \alpha)e^{y_1} + \alpha e^{x_2} + (1 - \alpha)e^{y_2},$$

i.e. $h(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) \leq \alpha h(\mathbf{x}) + (1 - \alpha) h(\mathbf{y})$.

Step 2 (push through g). Since g is *monotonically increasing*, applying it to the inequality of Step 1 preserves the direction:

$$F(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) = g(h(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y})) \leq g(\alpha h(\mathbf{x}) + (1 - \alpha) h(\mathbf{y})).$$

Step 3 (convexity of g). Since g is convex,

$$g(\alpha h(\mathbf{x}) + (1 - \alpha) h(\mathbf{y})) \leq \alpha g(h(\mathbf{x})) + (1 - \alpha) g(h(\mathbf{y})) = \alpha F(\mathbf{x}) + (1 - \alpha) F(\mathbf{y}).$$

Chaining Steps 2 and 3 gives

$$F(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) \leq \alpha F(\mathbf{x}) + (1 - \alpha) F(\mathbf{y}),$$

so F is convex. Note both hypotheses on g are used: monotonicity to transfer the inner inequality, and convexity to split the outer one.

Assignment 5 Let $P = \{(1, 2)^T, (-1, 0.5)^T, (0, 0.4)^T, (0.5, -0.5)^T, (-0.5, 1.5)^T\}$ and let $H = \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x}^T \mathbf{w} + b = 0\}$ with $\mathbf{w} = (1, -1)^T$ and $b = 0.5$. The hyperplane H splits P into two groups. Specify the groups and explain how they are obtained.

Solution. A point lies on one side of H according to the *sign* of the signed value $s(\mathbf{x}) = \mathbf{x}^T \mathbf{w} + b = x_1 - x_2 + 0.5$. Evaluating:

$\mathbf{x}$	$s(\mathbf{x}) = x_1 - x_2 + 0.5$	side
$(1, 2)$	$1 - 2 + 0.5 = -0.5$	—
$(-1, 0.5)$	$-1 - 0.5 + 0.5 = -1.0$	—
$(0, 0.4)$	$0 - 0.4 + 0.5 = +0.1$	+
$(0.5, -0.5)$	$0.5 + 0.5 + 0.5 = +1.5$	+
$(-0.5, 1.5)$	$-0.5 - 1.5 + 0.5 = -1.5$	—

Positive side ($s > 0$): $\{(0, 0.4)^T, (0.5, -0.5)^T\}$.

Negative side ($s < 0$): $\{(1, 2)^T, (-1, 0.5)^T, (-0.5, 1.5)^T\}$.

How obtained. The hyperplane H is the level set $s(\mathbf{x}) = 0$; it divides $\mathbb{R}^2$ into the half-space where $s(\mathbf{x}) > 0$ and the one where $s(\mathbf{x}) < 0$. Plugging each point into s and reading off the sign therefore tells us which side it falls on. (No point gives $s = 0$, so none lies exactly on H .)

Assignment 6 Consider $\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x})$ with $f(\mathbf{x}) = x_1^2 + 2x_2^2 - x_1x_2 + 1$. Choose a suitable step size $\eta > 0$ and run four steps of gradient descent from $\mathbf{x}_0 = (1, 1)^T$; report the result. Discuss the choice of η .

Solution. Gradient and minimizer.

$$\nabla f(\mathbf{x}) = \begin{pmatrix} 2x_1 - x_2 \\ 4x_2 - x_1 \end{pmatrix}, \quad \nabla^2 f = \begin{pmatrix} 2 & -1 \\ -1 & 4 \end{pmatrix} =: H.$$

Setting $\nabla f = \mathbf{0}$ gives $x_2 = 2x_1$ and $4x_2 = x_1$, hence the unique minimizer $\mathbf{x}^* = (0, 0)^T$ with $f(\mathbf{x}^*) = 1$. Since $H \succ 0$ (eigenvalues $\lambda = 3 \pm \sqrt{2}$, i.e. $\lambda_{\min} \approx 1.586$, $\lambda_{\max} \approx 4.414$), f is strictly convex and gradient descent converges to $\mathbf{x}^*$ for any $0 < \eta < 2/\lambda_{\max}$.

Choice of step size. We take $\eta = 0.2$, which is safely below $2/\lambda_{\max} \approx 0.453$, guaranteeing monotone convergence. The update is $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$:

k	$\mathbf{x}_k$	$\nabla f(\mathbf{x}_k)$	$f(\mathbf{x}_k)$
0	(1, 1)	(1, 3)	3.000
1	(0.800, 0.400)	(1.200, 0.800)	1.480
2	(0.560, 0.240)	(0.880, 0.400)	1.198
3	(0.384, 0.160)	(0.608, 0.256)	1.090
4	(0.2624, 0.1088)	(0.6080, 0.2560) [†]	1.064

([†] shown for completeness; not applied.) After four steps,

$$\boxed{\mathbf{x}_4 \approx (0.262, 0.109)^T, \quad f(\mathbf{x}_4) \approx 1.064}$$

which is converging toward the optimum $\mathbf{x}^* = (0, 0)^T$, $f^* = 1$.

Discussion of η . For a quadratic with Hessian H , gradient descent contracts the error by $\mathbf{x}_{k+1} - \mathbf{x}^* = (I - \eta H)(\mathbf{x}_k - \mathbf{x}^*)$, so it converges iff $|1 - \eta\lambda| < 1$ for every eigenvalue, i.e. $0 < \eta < 2/\lambda_{\max} \approx 0.453$. Too large an η (≥ 0.453) diverges or oscillates; too small an η converges slowly. The step size that minimizes the worst-case contraction factor is $\eta^* = \frac{2}{\lambda_{\min} + \lambda_{\max}} = \frac{2}{6} = \frac{1}{3}$, giving the optimal rate $\frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\sqrt{2}}{3} \approx 0.47$ per step; our choice $\eta = 0.2$ is a safe, conservative value in this admissible range.

AI usage report. I used an AI assistant (Claude) to help organize the writeup, cross-check the arithmetic of the Count-min sketch (Problem 1) and the four gradient-descent iterations (Problem 6), and to verify the algebra of the Perceptron mistake-bound derivation (Problem 3) and the convexity arguments (Problem 4). I designed and verified the Problem 2 counterexample and confirmed all final numerical answers independently.